neokim

6 AI AGENTIC WORKFLOW PATTERNS

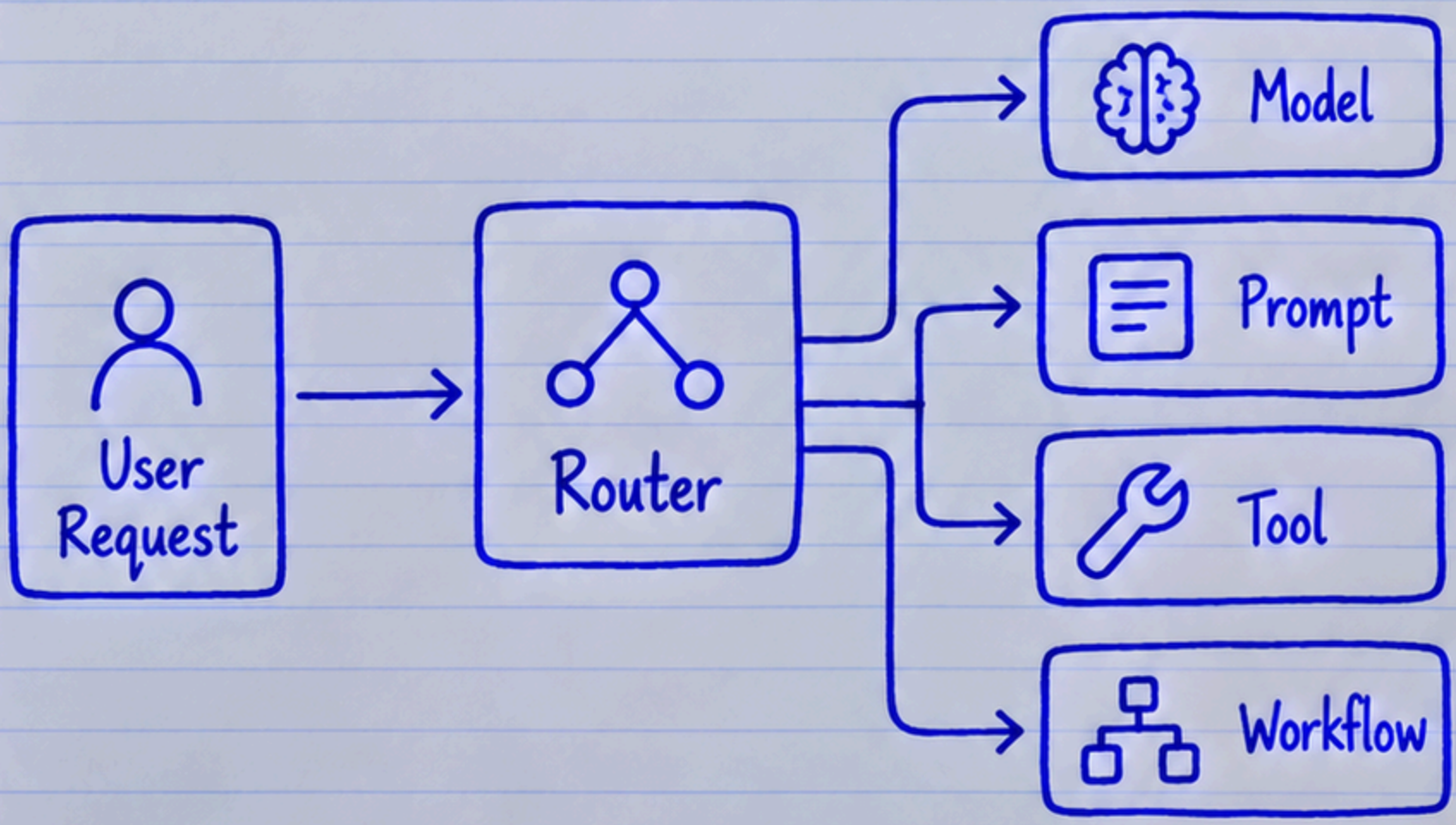
1/ Routing

Agent's request-routing workflow.

Classifies input & sends it to the appropriate model, prompt, tool, or workflow.

Different requests use different routes.

Failed routing or model calls may need retries and/or fallback handling.



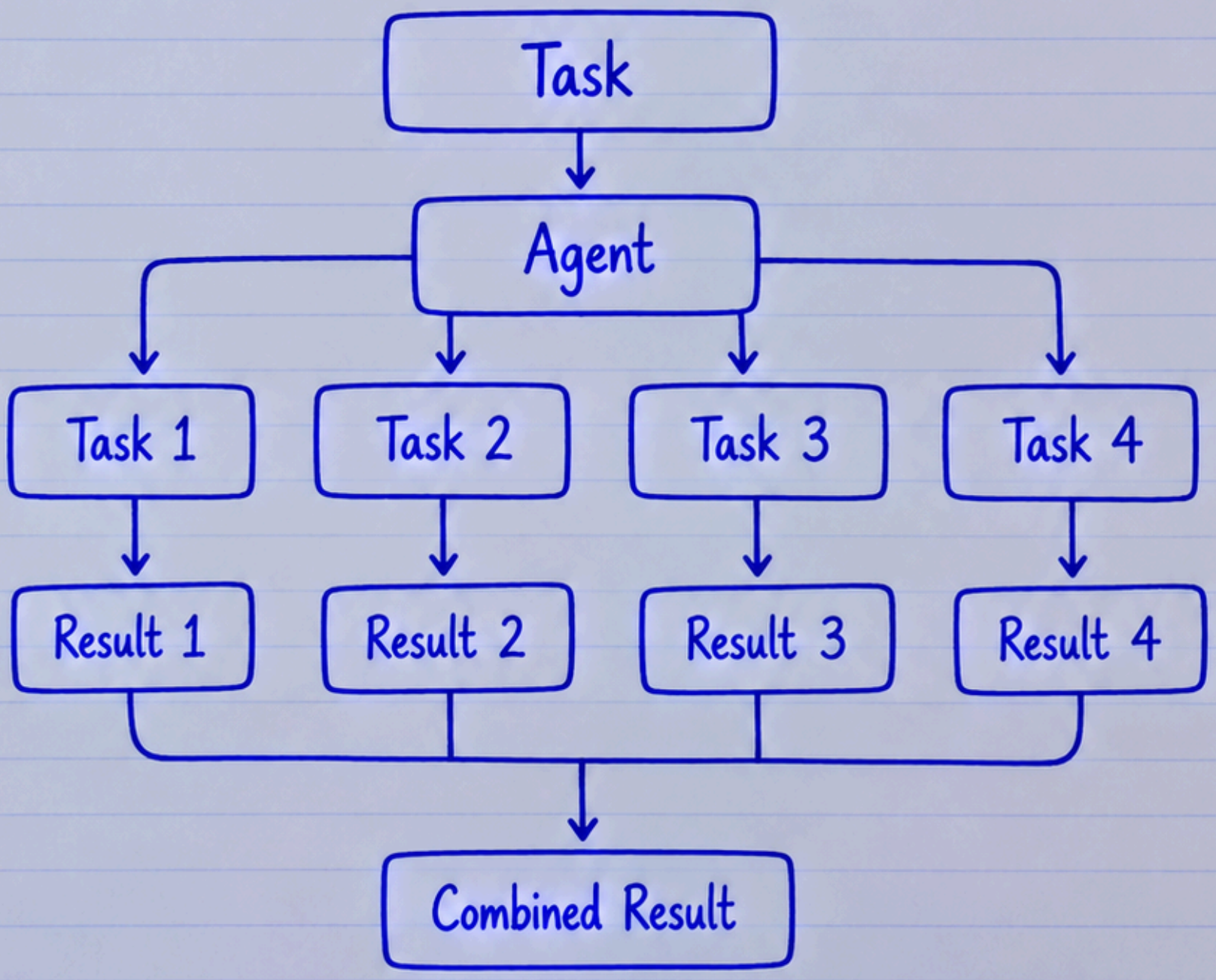
2/ Parallelization

Agent's parallel execution workflow.

Runs many predefined tasks simultaneously, then combines their results.

Sectioning runs different tasks in parallel, while voting runs same task many times.

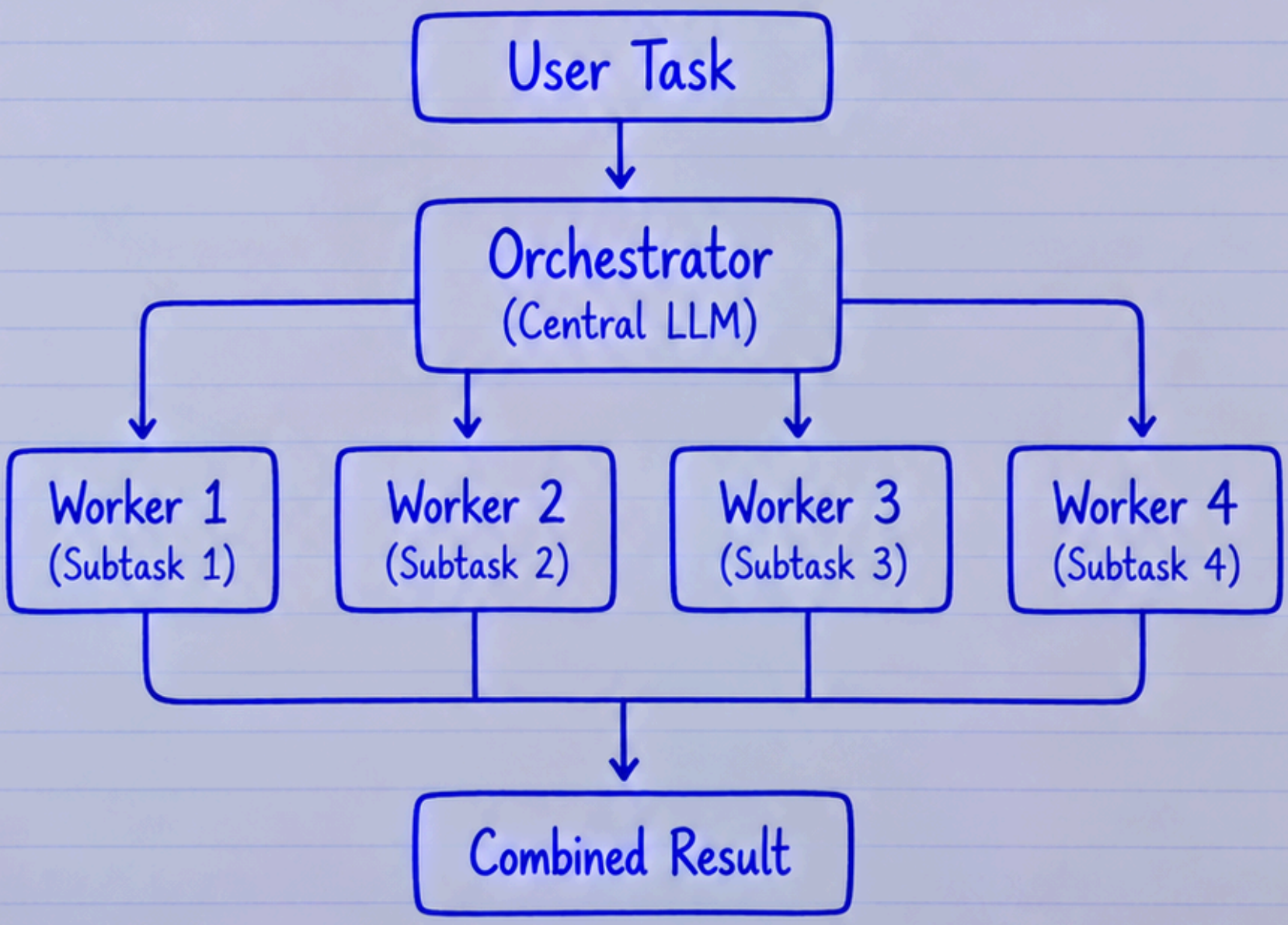
Parallel tasks could fail independently, so you need selective retries & partial failure handling.



3/ Orchestrator-Workers

Agent's task delegation workflow.
A central LLM breaks a task into subtasks,
assigns them to workers & combines their results.
The orchestrator determines which subtasks are
needed dynamically during execution.

Worker failures might leave tasks partially
completed, so you need recovery mechanisms.

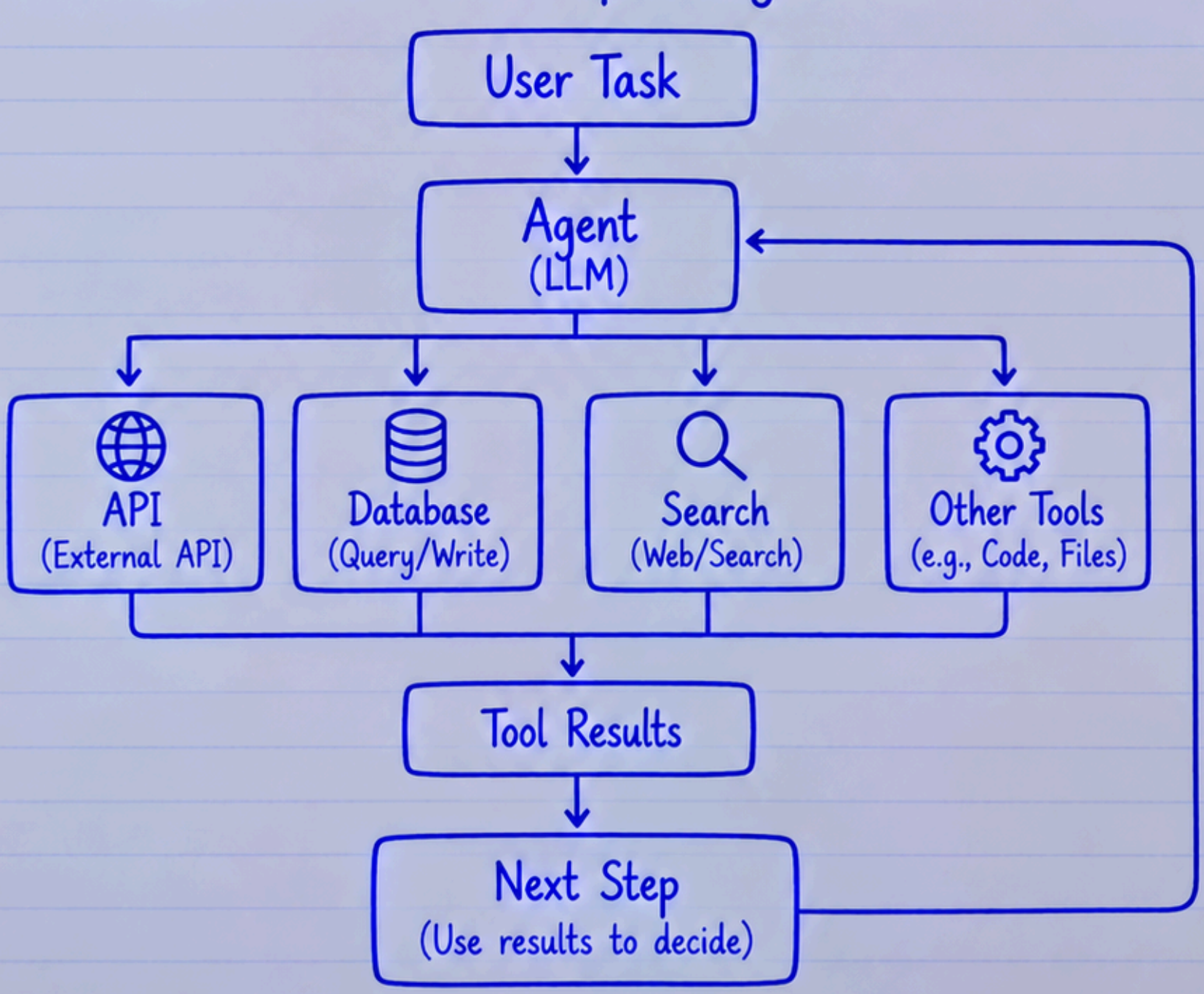


4/ Tool Use

Agent's tool execution workflow.

An LLM selects & calls external tools, such as APIs, databases & search, to complete a task. Tool results give the agent information it could use to decide what to do next.

Tool calls could fail/timeout, so you need bounded retries & idempotency.

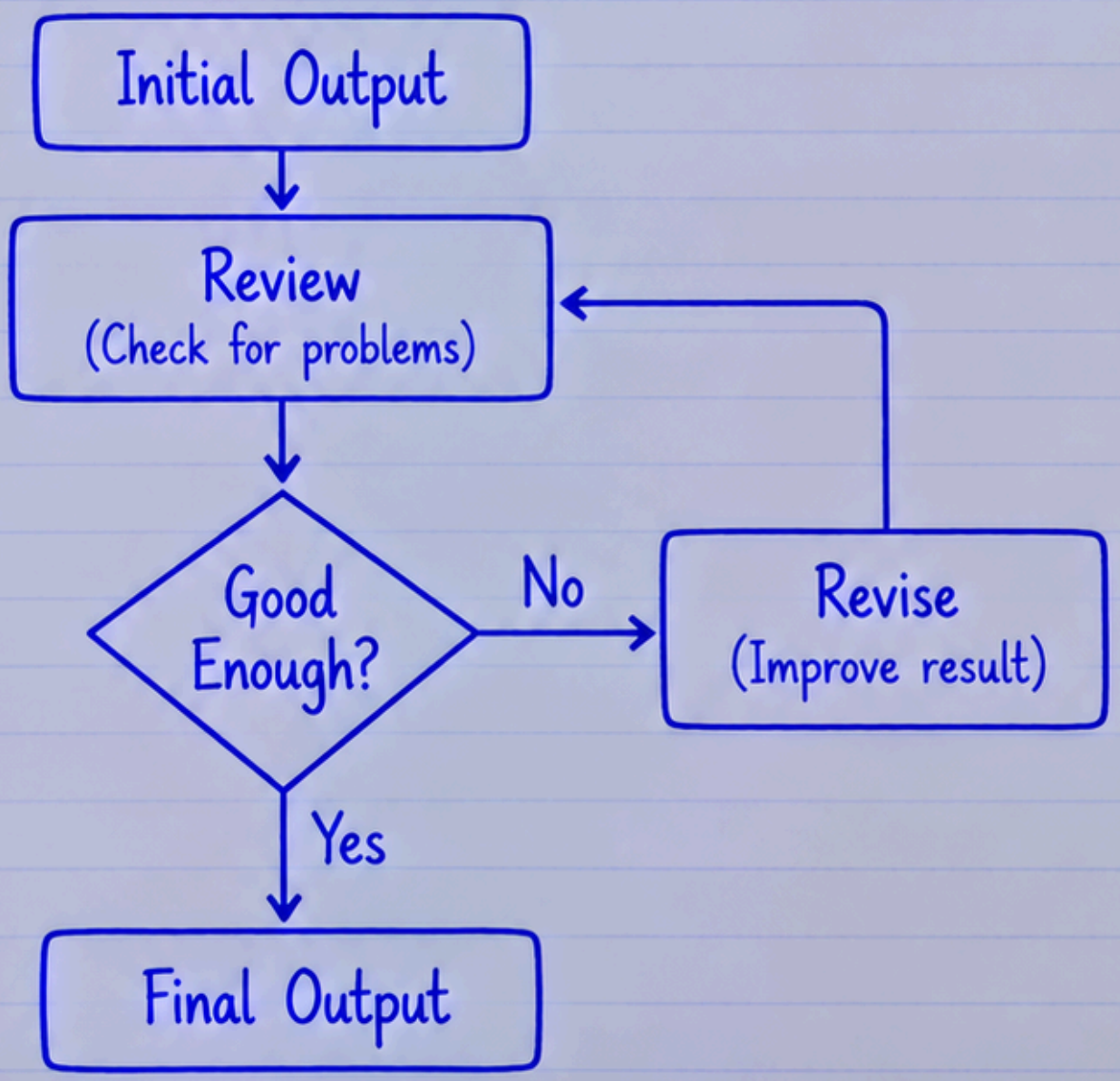


5/ Reflection

Agent's self-review workflow.

An LLM reviews its own output, identifies problems & revises result to improve quality. The agent repeats this process until it meets a quality threshold or a stopping limit.

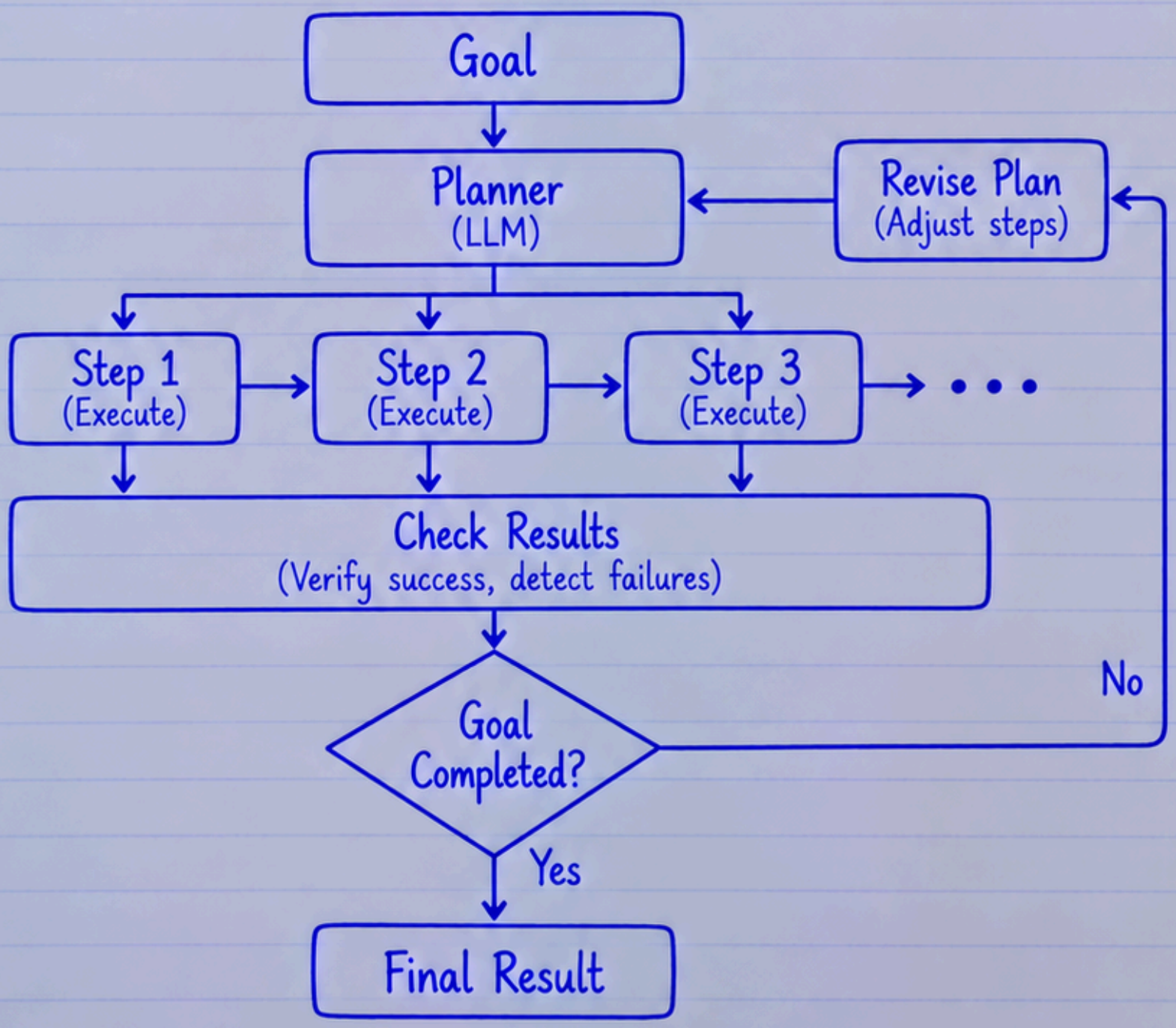
Reflection could get stuck in loops. So you need iteration limits & saved progress for recovery.



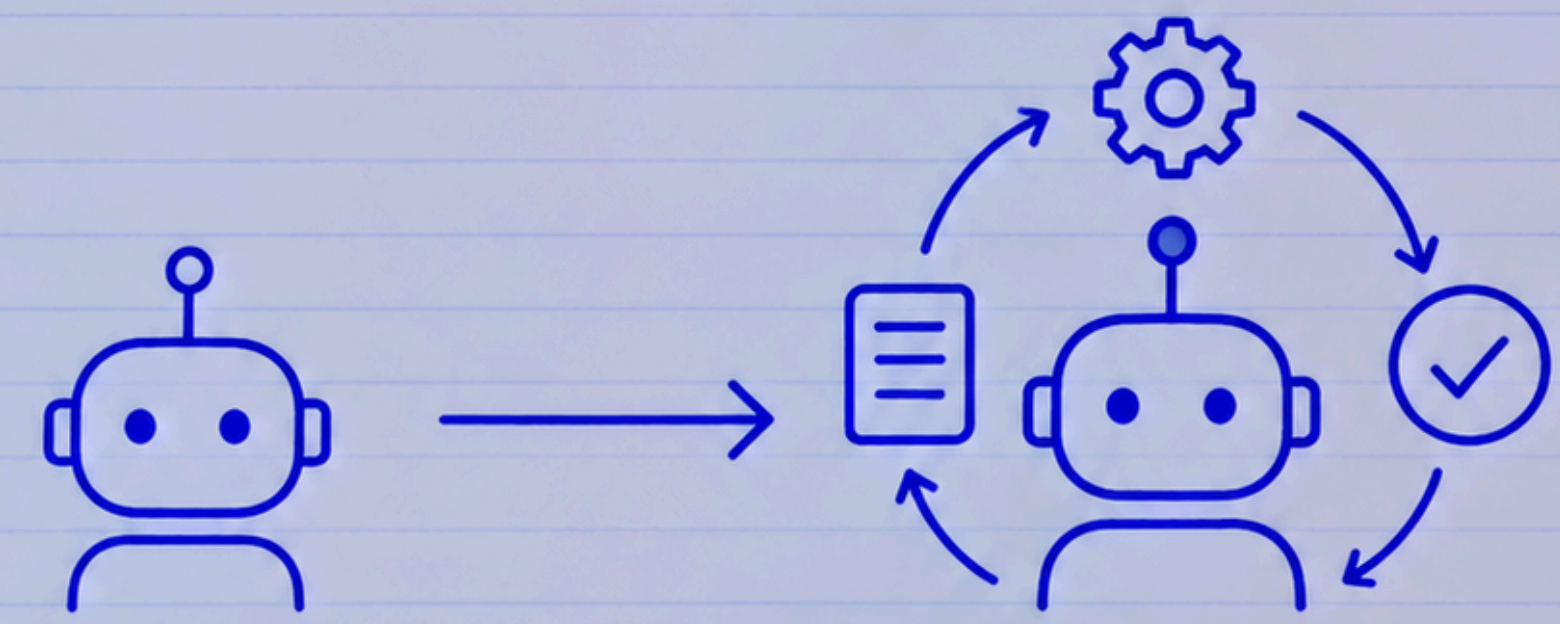
6/ Planning

Agent's task planning workflow.
An LLM breaks a complex goal into steps,
executes them & revises plan when needed.
The agent manages dependencies between steps
& adjusts its plan when execution fails.

Later steps may fail, so you need checkpoints &
recovery mechanisms to resume execution.



Master these & you go from building basic AI chatbots to building AI agents that plan, execute tasks & recover from failures.



RT 🔁 if this was useful.